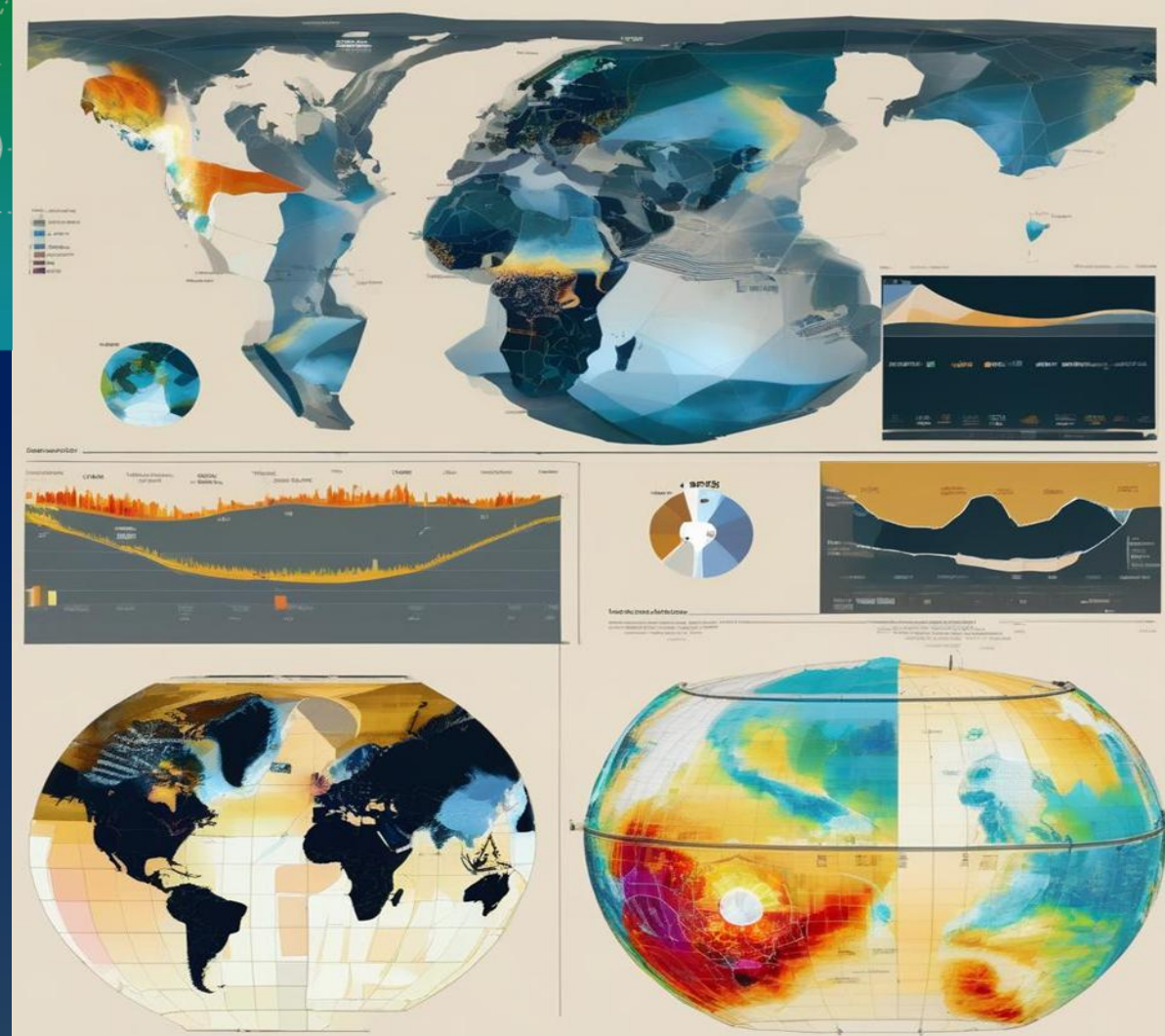




Aden (Yemen), 2 Dec - 4 Dec., 2025

Session IV. Exploiting CAVA data



Jane Mugo: jane.mugo@fao.org

Climate Risk Specialist

**Office of Climate Change, Biodiversity and Environment
(OCB, FAO)**

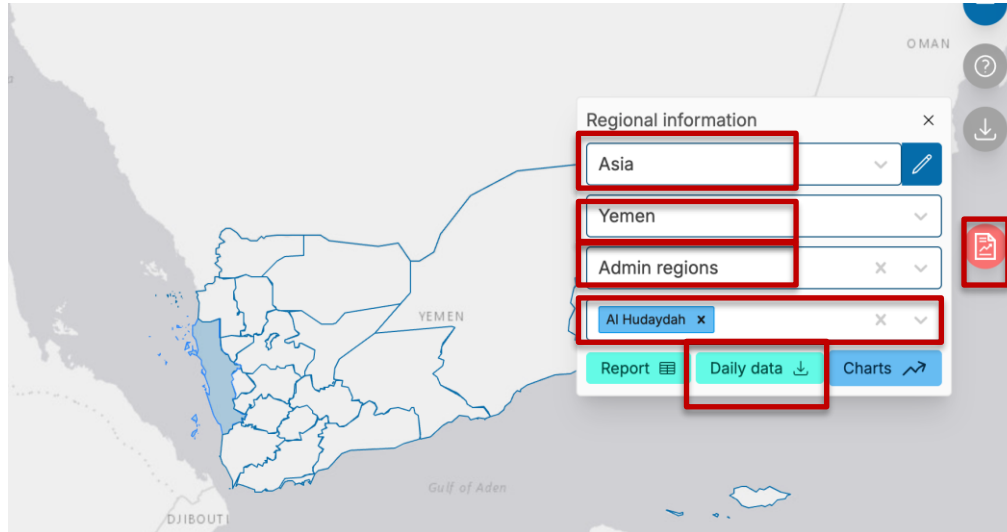
Presentation plan: Session V

1. Download daily climate data
2. Climate data visualization
3. Group exercises

1. Download daily climate data

CAVA: Download daily climate data

Step 1: Open CAVA



Step 2: Select the continent of interest (**Asia**)

Step 3: Select the country of interest (**Yemen**)

Step 4: Select the administrative boundaries (**Al Hudaydah**)

Step 5: Click on **DAILY DATA**

<https://cavaplatform.com/>

CAVA: Download daily climate data

The data download feature allows users to receive bias-corrected daily climate data for all CORDEX-CORE models and RCPs via email. The data is suitable for downstream analyses (e.g., crop models).

Fill Get data report

GET DATA

The download data feature allows users to receive bias-corrected daily climate data for all CORDEX-CORE models and RCPs by email. The data is suitable to be used for downstream analyses (e.g. crop models)

Email*

andrea.setti@fao.org

Region name*

Al Hudaydah

Region WKT*

MULTIPOLYGON(((42.73470067000005 13.604846509000026,42.71752431700003 13.603

Domain*

Domain identifies regions in the world according to CORDEX nomenclature. Click [here](#) for more info

Select...

VARIABLES

Variables will be bias-corrected with the empirical quantile mapping using W5E5 as observational dataset

tasmax x tasmin x pr x sfcWind x hurs x rsds x

End year*

In CMIP5, future simulations start in 2006 and run until the end of the century. By selecting an end year, you will receive bias-corrected daily climate data from 1980 up to the selected end year for all CORDEX-CORE models and RCPs

2100

Step 1: Enter your email address

Step 2: Enter the name of your region (**Al Hudaydah**)

Step 3: Select the area of interest (**WAS-22**)

Step 4: Select the climate variables of interest

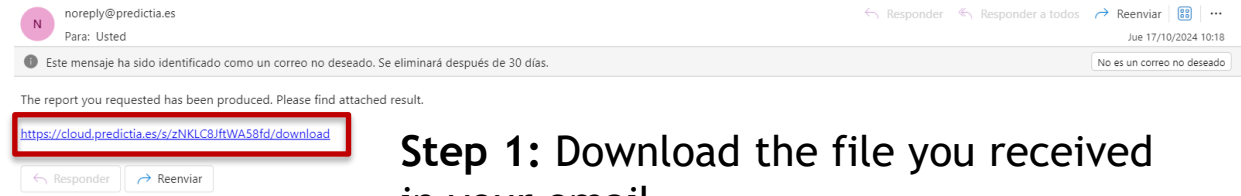
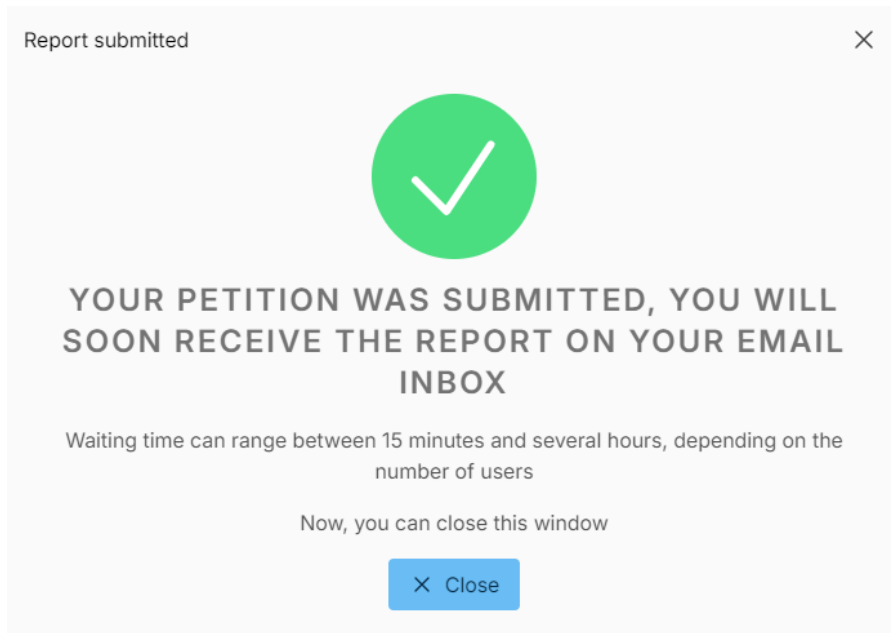
Tasmax = Maximum temperature (°C)
Tasmin = Minimum temperature (°C)
PR = Precipitation (mm)
scfWind = Average wind speed (m/s)
hurs = relative humidity (%)
Rds = Solar radiation (w/m2)

Step 5: Select the end year for which you want the daily data (**2100**)

Step 6: Click **SUBMIT** to obtain the data

CAVA: Download daily climate data

This process may take more than 15 minutes as we are downloading daily climate data (from 2006 to the end of the century) for 6 models (3 GCMs and 2 RCMs), 2 climate change scenarios (RCP 2.6 and 8.5), and 6 climate variables.



Step 1: Download the file you received in your email

CAVA: Download daily climate data

Step 1: Open the Excel file

Data

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	date	tmax (deg)	tmin (deg)	pr (mm)	sfcWind (m/)	hurs (%)	rsds (W/m^2)													
2	01/01/1980	32.87	20.083	0	0.949	66.91	223.471													
3	02/01/1980	33.861	20.964	0	0.975	61.759	233.738													
4	03/01/1980	33.999	21.396	0	1.076	55.51	243.712													
5	04/01/1980	34	20.89	0	1.172	64.016	225.518													
6	05/01/1980	33.17	22.09	0	1.327	72.958	194.691													
7	06/01/1980	32.077	22.754	6.789	1.065	76.971	175.78													
8	07/01/1980	33.675	22.197	0	1.282	71.956	191.687													
9	08/01/1980	32.597	22.595	0	1.066	75.471	195.454													
10	09/01/1980	32.37	22.906	0	1.03	74.587	183.414													
11	10/01/1980	33.879	21.498	0	1.102	75.316	195.885													
12	11/01/1980	33.735	22.386	0	1.226	72.139	209.683													
13	12/01/1980	33.259	21.886	0	1.351	75.394	200.454													
14	13/01/1980	33.441	21.921	0	1.468	76.898	192.659													
15	14/01/1980	32.953	23.166	0	1.315	75.671	186.833													
16	15/01/1980	33.632	22.591	0	1.212	75.059	177.422													
17	16/01/1980	31.494	22.609	0	1.212	79.214	156.827													
18	17/01/1980	34.283	22.407	0	1.336	73.04	190.704													
19	18/01/1980	32.906	22.322	0	1.385	73.474	196.034													
20	19/01/1980	32.924	23.167	0	1.214	75.026	203.786													
21	20/01/1980	34.675	22.978	0	1.489	71.413	192.171													
22	21/01/1980	33.298	22.565	0	1.29	73.635	212.642													
23	22/01/1980	33.912	23.018	0	1.535	72.668	217.526													
24	23/01/1980	35.08	23.552	0	1.326	71.695	206.329													
25	24/01/1980	34.669	22.729	0	1.397	68.788	213.487													
26	25/01/1980	35.212	21.962	0	1.277	68.298	223.526													
27	26/01/1980	33.736	22.672	0	1.315	64.447	241.628													
28	27/01/1980	33.954	22.937	0	1.239	64.419	245.04													
29	28/01/1980	34.333	22.622	0	1.15	64.207	229.861													
30	29/01/1980	35.562	22.063	0	1.441	64.51	241.997													
31	30/01/1980	33.898	22.723	0	1.24	68.117	227.412													
32	31/01/1980	36.611	23.023	0	1.272	61.608	246.648													
33	01/02/1980	35.881	22.937	0	1.446	65.375	251.717													
34	02/02/1980	35.571	22.869	0	1.4	62.639	249.544													
35	03/02/1980	34.694	23.374	0	1.104	57.806	265.04													
36	04/02/1980	34.981	20.97	0	1.124	60.447	245.963													
	< >	OBS	Model1_RCP26	Model2_RCP26	Model3_RCP26	Model4_RCP26	Model5_RCP26	Model6_RCP26	Model1_RCP85	Model2_RCP85	Model3_RCP85	Model4_RCP85	Model5_RCP85	Model6_RCP85						

Historical data
(W5E5)

Climate projections (3 GCMs x 2 RCMs x 2 RCPs)

2. Climate data visualization

CAVA: climate data visualization

Q Only selected Sync layers



Step 1: Click on **CLIMATE VARIABLES**

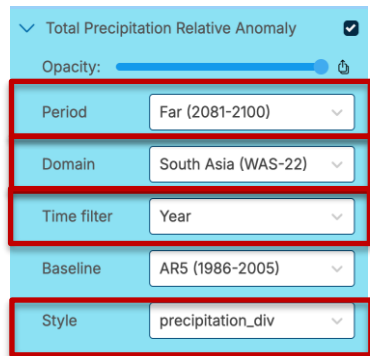
Step 2: Click on **PRECIPITATION**

Step 3: Click on **PROJECTIONS (CORDEX)**

Step 4: Click on **RCP 8.5**

Step 5: Click on **TOTAL PRECIPITATION
RELATIVE ANOMALY**

Repeat steps 1 to 9 for total precipitation, but this time under RCP 2.6



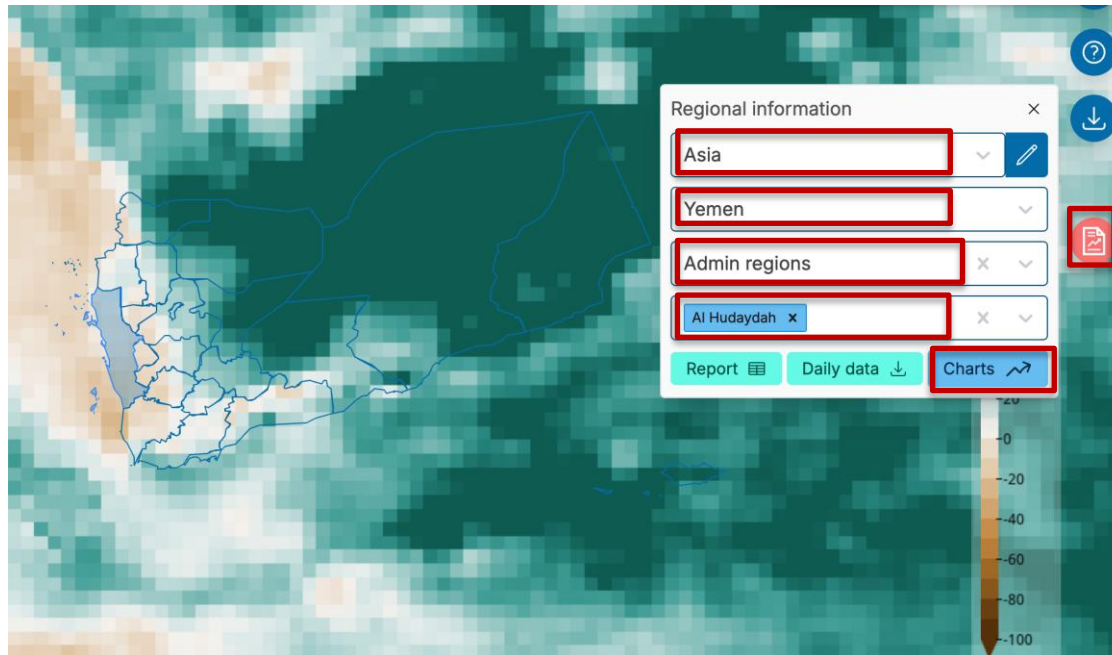
Step 6: Select the period of interest in the future (**e.g., 2081-2100**)

Step 7: Choose the area of interest (**WAS-22**)

Step 8: Choose the period of the year of interest (**YEAR**)

Step 9: Choose the style (**PRECIPITATION_DIV**)

CAVA: climate data visualization



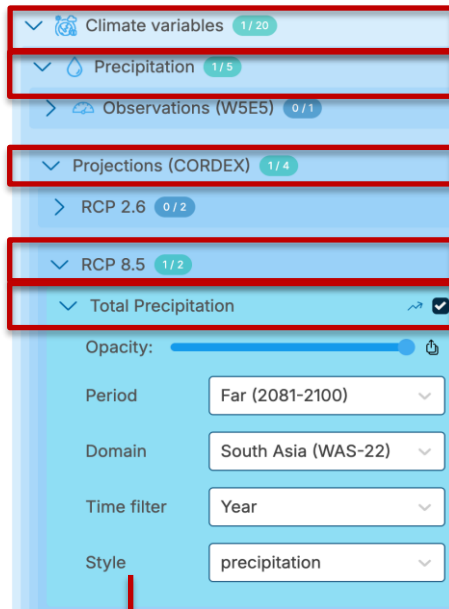
Step 1: Select the continent of interest (**Asia**)

Step 2: Select the country of interest (**Yemen**)

Step 3: Select the administrative boundaries (**Al Hudaydah**)

Step 4: Click on **CHARTS**

CAVA: climate data visualization



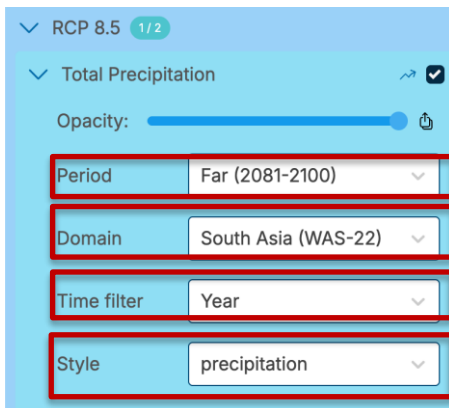
Step 1: Click on **CLIMATE VARIABLES**

Step 2: Click on **PRECIPITATION**

Step 3: Click on **PROJECTIONS (CORDEX)**

Step 4: Click on **RCP 8.5**

Step 5: Click on **TOTAL PRECIPITATION**



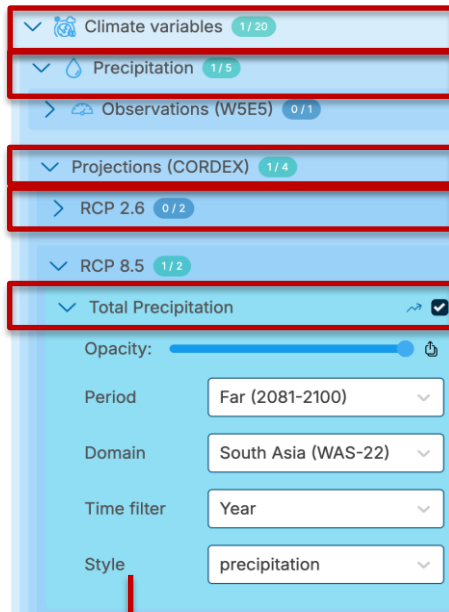
Step 6: Select the period of interest in the future (**e.g., 2081-2100**)

Step 7: Choose the area of interest (**WAS-22**)

Step 8: Choose the period of the year of interest (**YEAR**)

Step 9: Choose the style (**PRECIPITATION**)

CAVA: climate data visualization



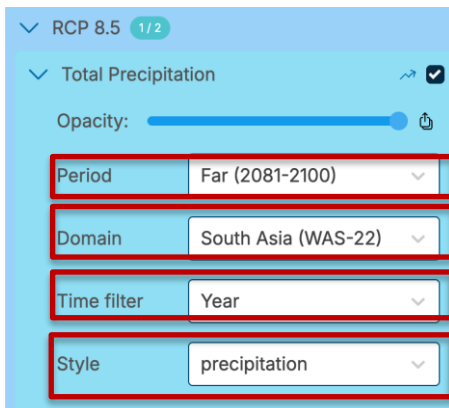
Step 1: Click on **CLIMATE VARIABLES**

Step 2: Click on **PRECIPITATION**

Step 3: Click on **PROJECTIONS (CORDEX)**

Step 4: Click on **RCP 2.6**

Step 5: Click on **TOTAL PRECIPITATION**



Step 6: Select the period of interest in the future (**e.g., 2081-2100**)

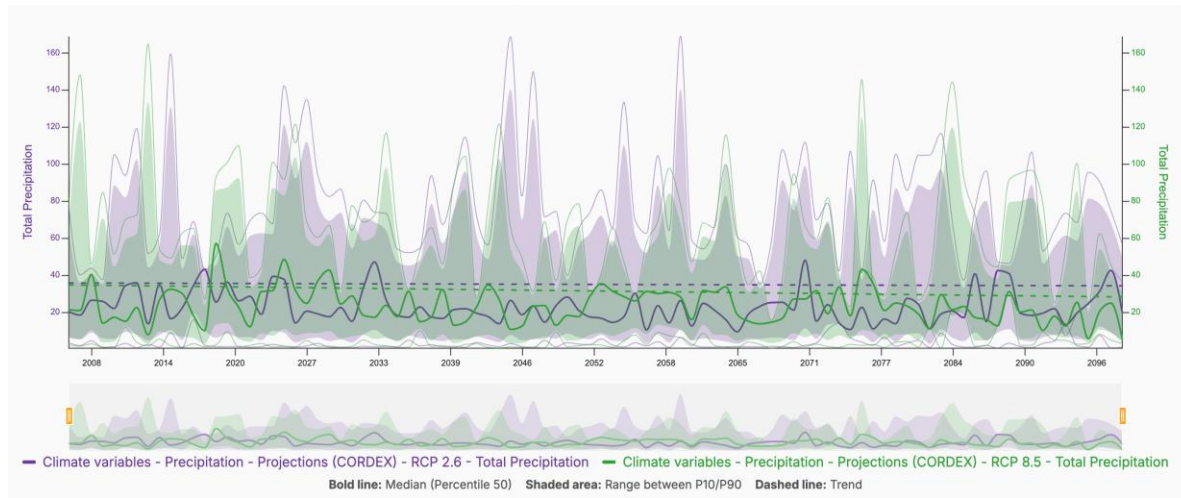
Step 7: Choose the area of interest (**WAS-22**)

Step 8: Choose the period of the year of interest (**YEAR**)

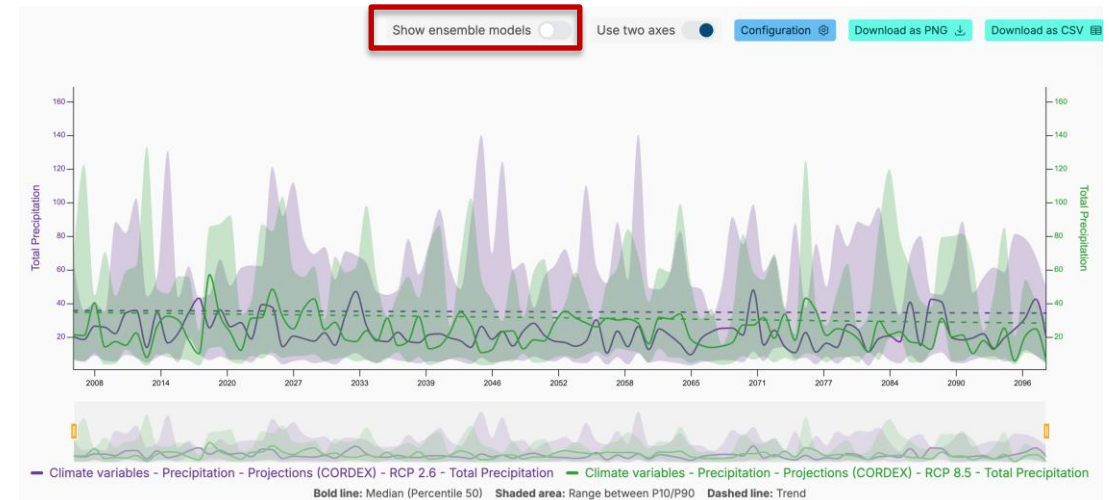
Step 9: Choose the style (**PRECIPITATION**)

CAVA: climate data visualization

We now have data for 3 GCMs, 2 RCMs, and 2 RCPs (2.6 and 8.5) for the period 2006-2099



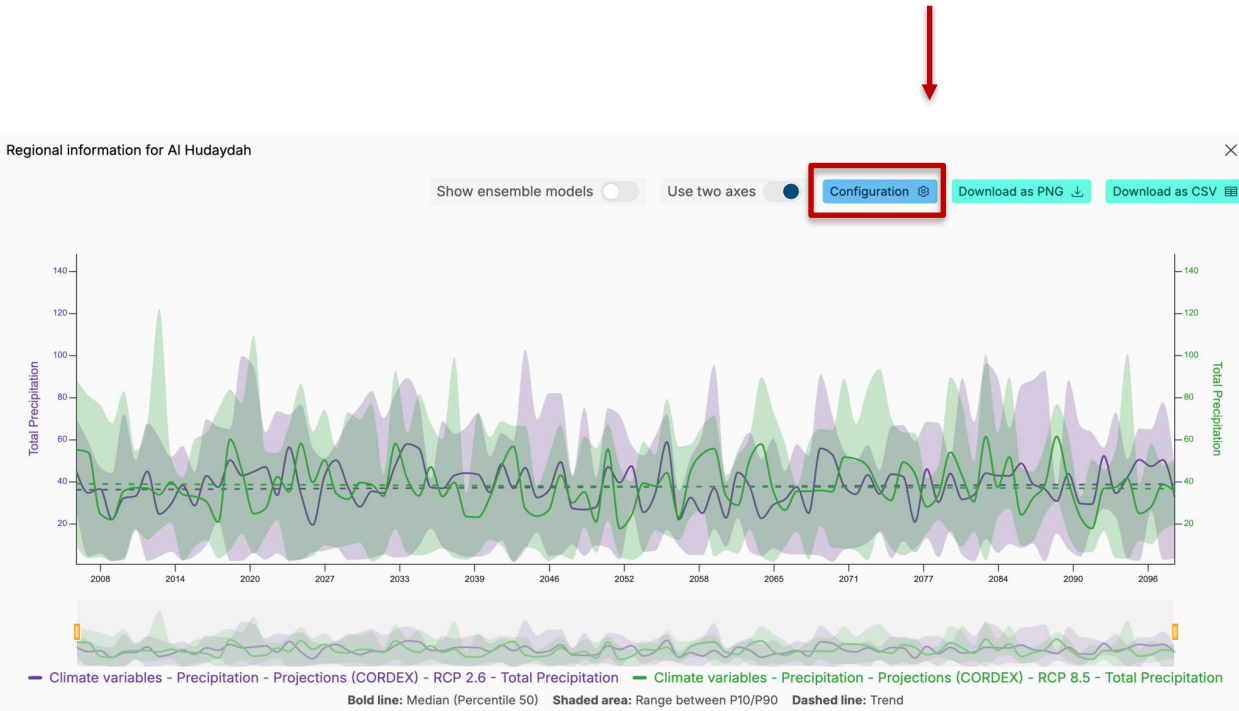
Step 1: Uncheck the ensemble model function



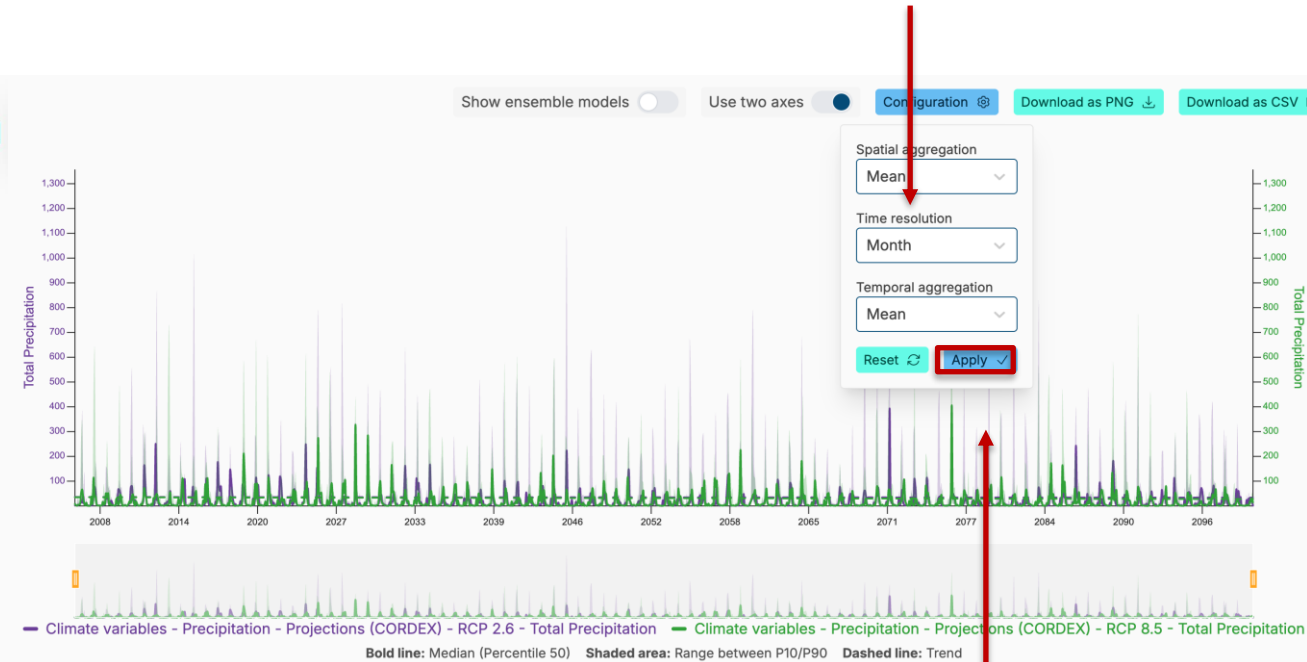
Remember that CAVA has averaged (ensemble) all pixels in the Al Hudaydah region

CAVA: climate data visualization

Step 1: Click on **CONFIGURATION**



Step 2: Select **MONTH**



Step 3: Click **APPLY**

CAVA: climate data visualization

Step 1: Click **DOWNLOAD AS PNG**

Step 2: Click **DOWNLOAD AS CSV**

Regional information for Al Hudaydah

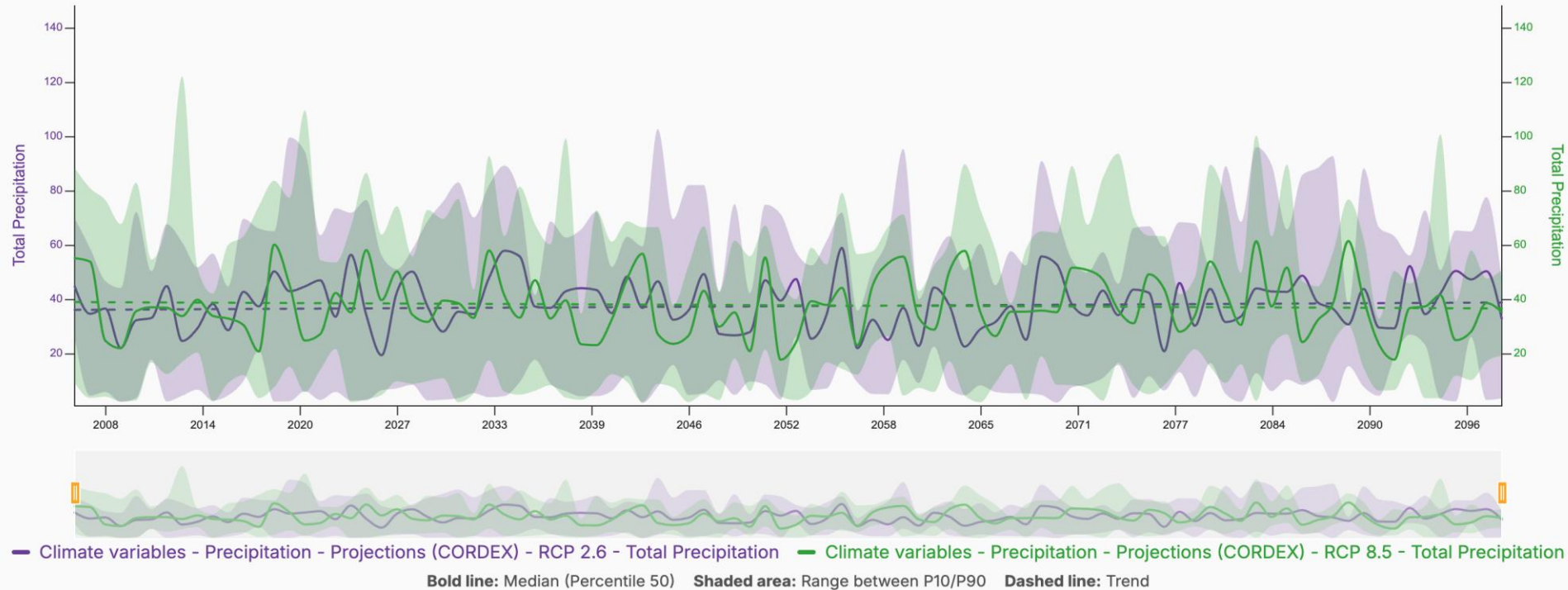
Show ensemble models ☐

Use two axes ☒

Configuration 

Download as PNG 

Download as CSV 



The end!
